

Food Safety Management System - Updated End User Operations

User Roles & Permissions

Admin:

- Full access to all features
- Create/manage users
- Create/edit/delete products, categories, batches, storage, inspections
- View-only access to Risk Analysis and Alerts (automatic)
- Access to AI Chatbot for intelligent assistance

Quality Control Officer:

- Create/edit products, categories, batches, storage, inspections
- View-only access to Risk Analysis and Alerts (automatic)
- Access to AI Chatbot for intelligent assistance
- Cannot manage users

Production Staff:

- Create/edit products, categories, batches, storage
- View-only access to Risk Analysis and Alerts (automatic)
- Access to AI Chatbot for intelligent assistance
- Cannot perform inspections or manage users

Manager:

- View all data (read-only dashboard access)
- Monitor safety status across all modules
- Access to AI Chatbot for intelligent assistance

System Workflow (Fully Automated + AI Assistant)

1. Login

- Users authenticate with email/password
- Enhanced login page with system information
- System assigns role-based permissions
- Redirects to dashboard based on role

2. Dashboard

- Overview of key metrics:
 - Total Products
 - Total Batches
 - Unsafe Batches
- Quick access to all modules via sidebar

3. AI Chatbot (NEW)

- **Floating chat widget** available on all pages
- **Natural language queries** - ask questions in plain English
- **System Analysis:**
 - "Analyze trends" - View risk patterns
 - "Predict risks" - AI-powered risk forecasting
 - "Generate report" - Automatic safety reports
 - "Proactive suggestions" - Get recommendations
- **Data Queries:**
 - "How many products/batches?"
 - "Show unsafe batches"
 - "Risk for batch X"
 - "Show alerts" or "unread alerts"
 - "Storage conditions"
 - "Inspections"
- **Regulations:**
 - "Temperature regulations"
 - "Humidity requirements"
 - "Expiry guidelines"
 - "Food safety standards"

4. Products Management

- **Create:** Add new products with name and category
- **View:** List all products in table format

- **Edit/Delete:** Manage product details (Admin/QC Officer/Production Staff)

5. Categories Management

- **Create:** Define product categories
- **View:** List all categories
- **Edit/Delete:** Manage categories (Admin/QC Officer/Production Staff)

6. Batches Management

- **Create:** Add production batches with:
 - Product selection
 - Batch number
 - Production date
 - Expiry date
 - Quantity
- **View/Edit/Delete:** Manage batch information

7. Storage Conditions (Triggers Auto Risk Analysis)

- **Create:** Record storage data for batches:
 - Temperature (°C)
 - Humidity (%)
 - Storage type
- **Automatic:** When storage is saved, AI immediately evaluates risk
- **View/Edit/Delete:** Manage storage records

8. Inspections

- **Create:** Log quality inspections:
 - Select batch
 - Inspector ID
 - Status (Passed/Failed)
 - Remarks
- **View:** Inspection history with status badges
- **Edit/Delete:** Manage inspection records (Admin/QC Officer)

9. Risk Analysis (Fully Automatic)

- **Automatic Generation:** Created when storage conditions are recorded
- **AI Evaluation:** Calculates risk based on:
 - Temperature (>35°C = high risk)
 - Humidity (>70% = high risk)
 - Expiry date (expired or near expiry = high risk)
- **View Only:** Read-only dashboard showing all risk analyses
- **No Manual Input:** Users cannot manually create or edit risk analysis

10. Alerts (Fully Automatic)

- **Automatic Generation:** Created when risk is Warning or Unsafe
- **Auto Messages:** System generates appropriate alert messages
- **View Only:** Read-only dashboard showing all alerts
- **Status Tracking:** Alerts marked as "Unread" initially
- **No Manual Input:** Users cannot manually create or edit alerts

Typical User Workflow

Production Staff:

1. Login → Create categories → Create products
2. Create batches for production
3. Record storage conditions (temperature, humidity)
4. **System automatically evaluates risk and generates alerts**
5. Use AI Chatbot to ask: "Show storage conditions" or "Predict risks"
6. Monitor batch status

Quality Control Officer:

1. Login → View batches → Perform inspections
2. Record storage conditions
3. **System automatically evaluates risk and generates alerts**
4. Use AI Chatbot to ask: "Analyze trends" or "Generate report"
5. Review automatic risk analysis and alerts
6. View inspection results

Admin:

1. Login → Create user accounts
2. Manage all system data
3. Monitor overall system health via dashboard
4. Use AI Chatbot to ask: "Proactive suggestions" or "Show unsafe batches"
5. View automatic risk analysis and alerts

Manager:

1. Login → View dashboard metrics
2. Monitor safety status across all batches
3. Use AI Chatbot to ask: "Generate report" or "Temperature regulations"
4. Review automatically generated alerts and risk analysis

Key System Features (Updated)

- **Fully Automated Risk Analysis:** No manual evaluation needed
- **Automatic Alert Generation:** Alerts created instantly for unsafe conditions
- **AI-Powered Chatbot:** Natural language assistance for all queries
- **Trend Analysis:** View risk patterns and distributions
- **Risk Prediction:** AI forecasting of potential risks
- **Automatic Reports:** Comprehensive safety reports on demand
- **Proactive Suggestions:** AI recommendations based on system data
- **Real-time Data:** All operations update immediately
- **Role-Based Access:** Users see only features they're permitted to use
- **AI-Powered Safety Scoring:** Intelligent evaluation based on multiple factors
- **Passive Monitoring:** Users only need to record storage conditions; system handles the rest
- **Responsive Design:** Works on desktop and mobile devices